

```

library(wordcloud)
library(topicmodels)
library(tidytext)
library(dplyr)

setwd("C:/Users/Emma/Documents/Senior Seminar Paper")
# set working directory
email_txt <- file.path("emails")
# chose folder with email text file

# create raw corpus, which will be preprocessed in a moment
email <- VCorpus(DirSource(email_txt))

# cleaning/ preprocessing

email <- tm_map(email, tolower)
email <- tm_map(email, removeWords, stopwords("english"))
email <- tm_map(email, removePunctuation)
email <- tm_map(email, removeNumbers)
email <- tm_map(email, stripWhitespace)
email <- tm_map(email, PlainTextDocument)

for (char in seq(email)){
  email[[char]] <- gsub("statgovgt", " ", email[[char]])
  email[[char]] <- gsub("â€", " ", email[[char]])
  email[[char]] <- gsub("@", " ", email[[char]])
  email[[char]] <- gsub("ltsullivanjj", " ", email[[char]])
  email[[char]] <- gsub("ltmlscd", " ", email[[char]])
  email[[char]] <- gsub("lthdrclint", " ", email[[char]])
  email[[char]] <- gsub("ltabedinh", " ", email[[char]])
  email[[char]] <- gsub("mailcomgt", " ", email[[char]])
  email[[char]] <- gsub("lthdr", " ", email[[char]])
  email[[char]] <- gsub("clintonemailcomgt", " ", email[[char]])
}

otherwords <- c("can", "well", "let", "best", "like", "now", "last", "sun", "still", "dont", "see", "next",
  "now", "mar", "also", "case", "subject", "unclassified", "department", "can", "state",
  "two", "one", "know", "doc", "date", "will", "made", "new",
  "just", "think", "want", "sent", "release", "part", "january", "july", "get", "original",
  "message", "said", "first", "sat", "sure", "whether", "even", "fri", "jan", "yet",
  "even", "feb", "day", "day", "got", "tue", "may", "thu", "wed", "mon", "seen",
  "oct", "statgov", "email", "tuesday", "december",
  "clintone", "sunday", "clinton", "friday", "really", "sid", "pis", "else", "way", "monday", "mar",
  "mailtohdr", "sep", "nov", "end", "aug", "lthdr", "since", "office", "dec", "h", "c", "cc", "b",
  "fw", "f", "gt", "pm", "re", "thx", "make", "thanks", "make", "known", "today", "thursday",
  "forward", "wednesday", "week", "full", "jiloty", "lthrodclintone", "us", "ltvermarr",
  "lrmchaleja", "ltslaughtera", "das", "gis", "dos", "s", "april", "february")

```

```
email <- tm_map(email, removeWords, otherwords)
```

```
email <- tm_map(email, stripWhitespace)  
email <- tm_map(email, PlainTextDocument)
```

DTM and Frequency Code, and Repeated with Name Removal

```
# dtm: each document is a row, and each term is a column  
dtm <- DocumentTermMatrix(email)  
  
frequency <- sort(colSums(as.matrix(dtm)), decreasing=TRUE)  
# adding up the number of times each term is used, and sorting based on frequency of usage  
  
names <- c("cheryl", "huma", "abedin", "mills", "jacob", "sullivan", "david",  
          "verma", "lauren", "valmoro", "lona", "mchale", "judith", "richard", "crowley",  
          "philip")  
email_nonames <- tm_map(email, removeWords, names)  
email_nonames <- tm_map(email_nonames, PlainTextDocument)  
  
# create new dtm and calculate new frequencies  
  
dtm_nonames <- DocumentTermMatrix(email_nonames)  
  
frequency_nonames <- sort(colSums(as.matrix(dtm_nonames)), decreasing=TRUE)
```

Ngrams Code

```
BigramTokenizer <- function(x)  
  unlist(lapply(ngrams(words(x), 2), paste, collapse = " "), use.names = FALSE)  
tdm.bigram <- TermDocumentMatrix(email, control = list(tokenize = BigramTokenizer))  
tdm.bigram  
  
freq <- sort(rowSums(as.matrix(tdm.bigram)), decreasing = TRUE)  
freq.df <- data.frame(word=names(freq), freq=freq)  
head(freq.df, 50)  
  
TrigramTokenizer <- function(x)  
  unlist(lapply(ngrams(words(x), 3), paste, collapse = " "), use.names = FALSE)  
tdm.trigram <- TermDocumentMatrix(email, control = list(tokenize = TrigramTokenizer))  
  
freq <- sort(rowSums(as.matrix(tdm.trigram)), decreasing = TRUE)  
freq.df <- data.frame(word=names(freq), freq=freq)  
head(freq.df, 50)
```

```

BigramTokenizer <- function(x)
  unlist(lapply(ngrams(words(x), 2), paste, collapse = " "), use.names = FALSE)
tdm.bigram <- TermDocumentMatrix(email_nonames, control = list(tokenize = BigramTokenizer))

freq_nonames <- sort(rowSums(as.matrix(tdm.bigram)),decreasing = TRUE)
freq.df <- data.frame(word=names(freq_nonames), freq=freq_nonames)
head(freq.df, 50)

TrigramTokenizer <- function(x)
  unlist(lapply(ngrams(words(x), 3), paste, collapse = " "), use.names = FALSE)
tdm.trigram <- TermDocumentMatrix(email_nonames, control = list(tokenize = TrigramTokenizer))

freq_nonames <- sort(rowSums(as.matrix(tdm.trigram)),decreasing = TRUE)
freq.df <- data.frame(word=names(freq_nonames), freq=freq_nonames)
head(freq.df, 50)

```

Topic Models

```

# 2 groups

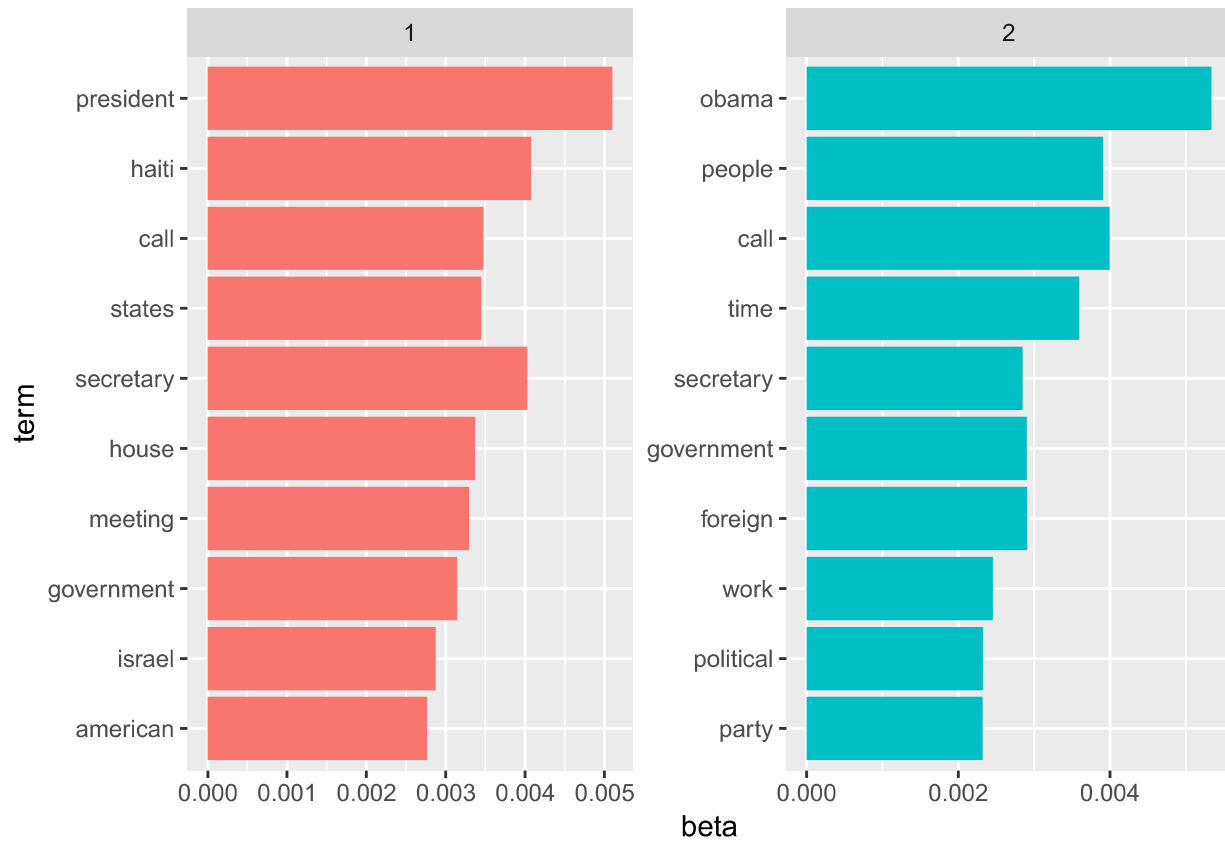
email_nonames_2topic <- LDA(dtm_nonames, k=2 )

topics2 <- tidy(email_nonames_2topic, matrix= "beta")

top_terms <- topics2 %>%
  group_by(topic)%>%
  top_n(10, beta) %>%
  ungroup()%>%
  arrange(topic, -beta)

top_terms %>%
  mutate(term = reorder(term, beta)) %>%
  ggplot(aes(term, beta, fill = factor(topic))) +
  geom_col(show.legend = FALSE) +
  facet_wrap(~ topic, scales = "free") +
  coord_flip()

```



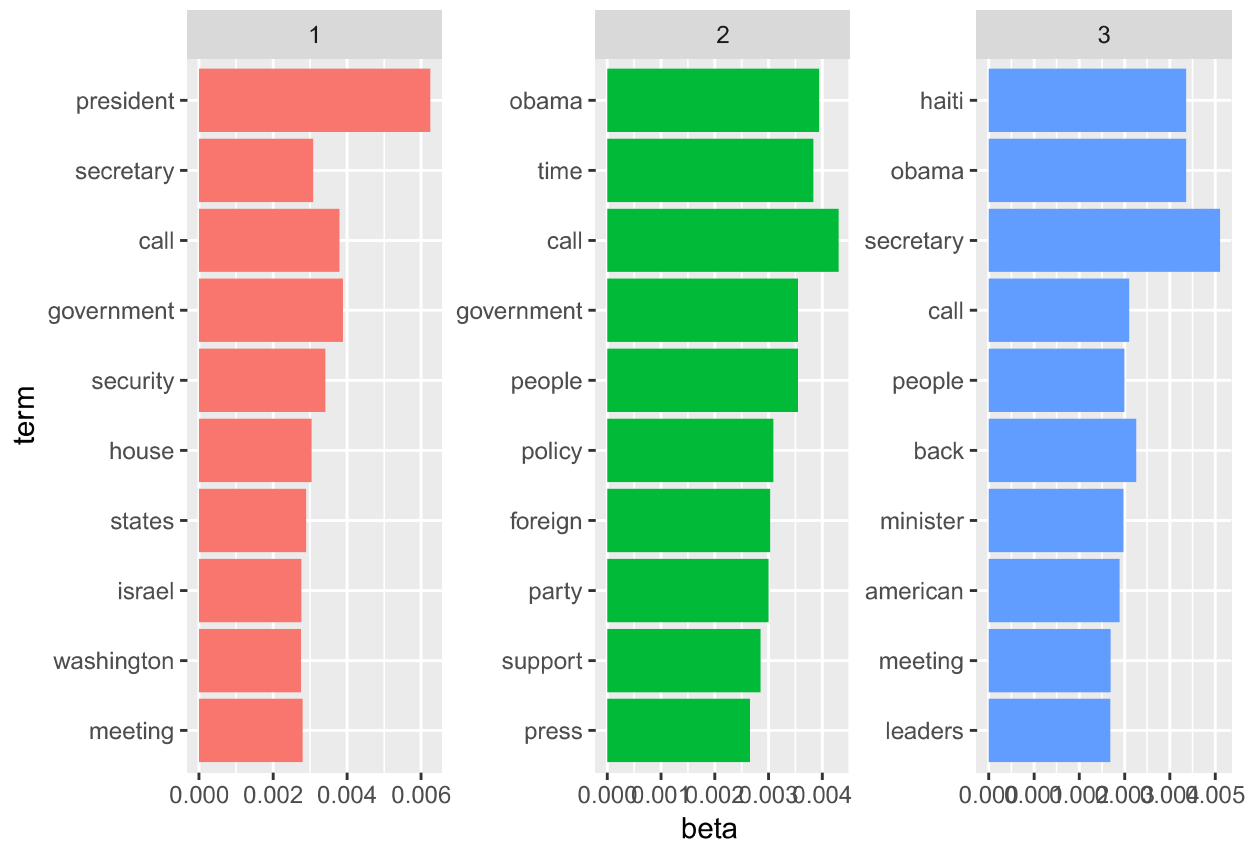
3 groups

```
email_nonames_3topic <- LDA(dtm_nonames, k=3 )

topics3 <- tidy(email_nonames_3topic, matrix= "beta")

top_terms <- topics3%>%
  group_by(topic)%>%
  top_n(10, beta) %>%
  ungroup()%>%
  arrange(topic, -beta)

top_terms %>%
  mutate(term = reorder(term, beta)) %>%
  ggplot(aes(term, beta, fill = factor(topic))) +
  geom_col(show.legend = FALSE) +
  facet_wrap(~ topic, scales = "free") +
  coord_flip()
```



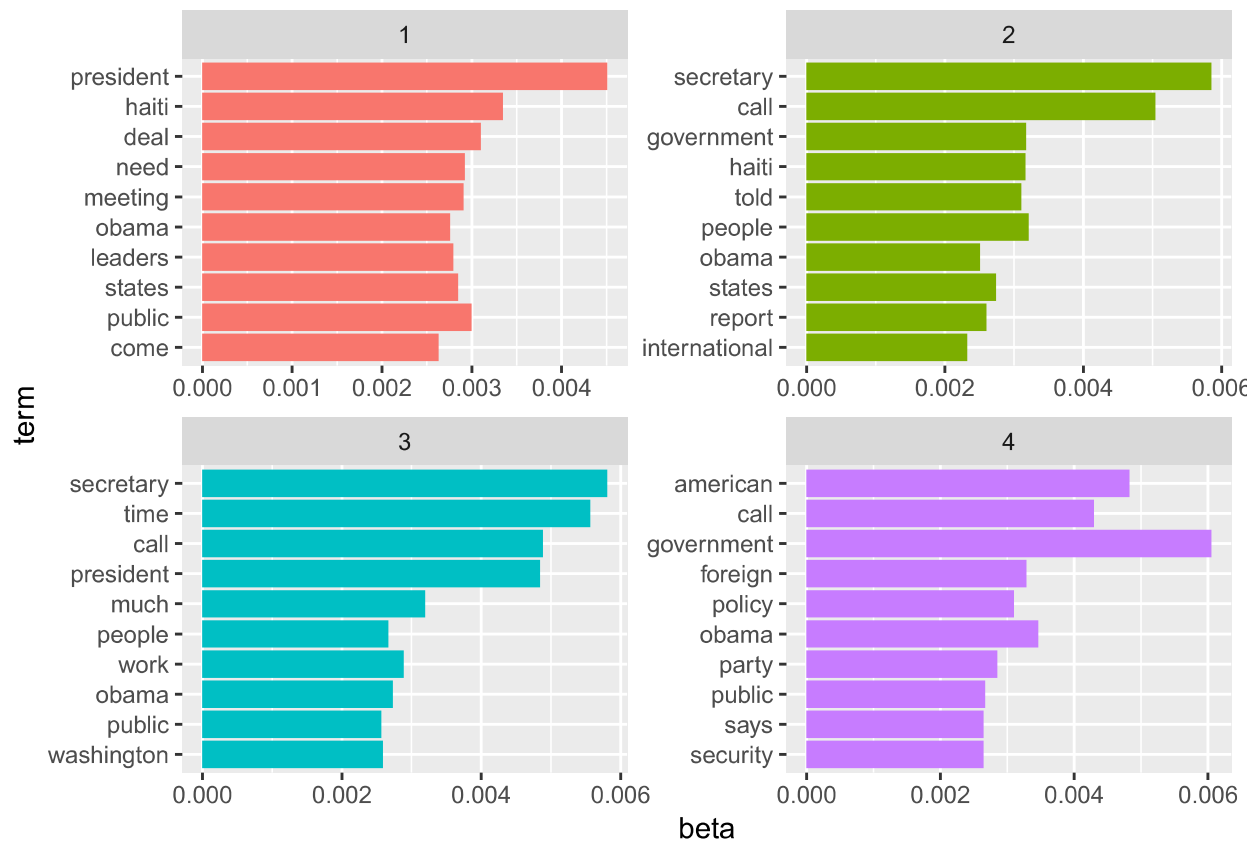
4 groups

```
email_nonames_4topic <- LDA(dtm_nonames, k=4 )

topics4 <- tidy(email_nonames_4topic, matrix= "beta")

top_terms <- topics4%>%
  group_by(topic)%>%
  top_n(10, beta) %>%
  ungroup()%>%
  arrange(topic, -beta)

top_terms %>%
  mutate(term = reorder(term, beta)) %>%
  ggplot(aes(term, beta, fill = factor(topic))) +
  geom_col(show.legend = FALSE) +
  facet_wrap(~ topic, scales = "free") +
  coord_flip()
```



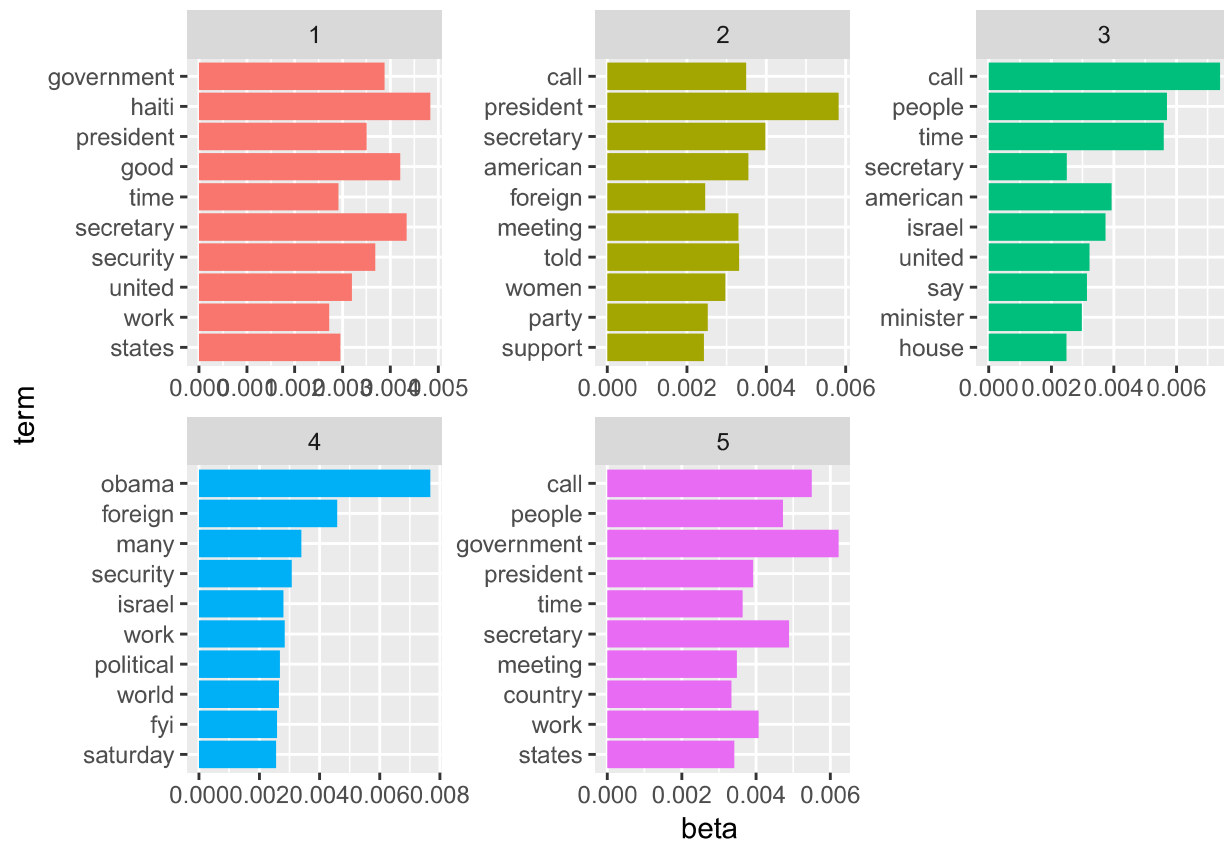
5 groups

```
email_nonames_5topic <- LDA(dtm_nonames, k=5 )

topics5 <- tidy(email_nonames_5topic, matrix= "beta")

top_terms <- topics5%>%
  group_by(topic)%>%
  top_n(10, beta) %>%
  ungroup()%>%
  arrange(topic, -beta)

top_terms %>%
  mutate(term = reorder(term, beta)) %>%
  ggplot(aes(term, beta, fill = factor(topic))) +
  geom_col(show.legend = FALSE) +
  facet_wrap(~ topic, scales = "free") +
  coord_flip()
```



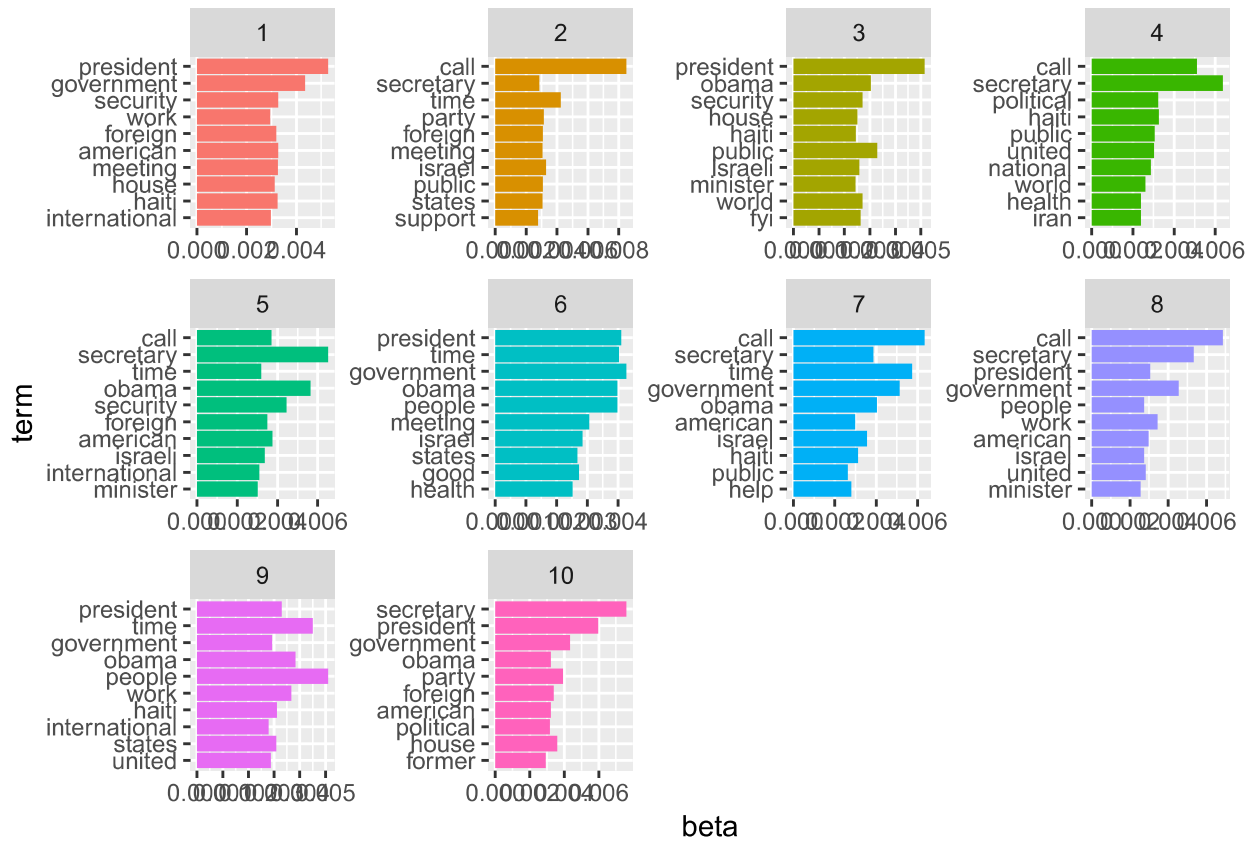
```
# 10

email_nonames_10topic <- LDA(dtm_nonames, k=10)

topics10 <- tidy(email_nonames_10topic, matrix= "beta")

top_terms <- topics10%>%
  group_by(topic)%>%
  top_n(10, beta) %>%
  ungroup()%>%
  arrange(topic, -beta)

top_terms %>%
  mutate(term = reorder(term, beta)) %>%
  ggplot(aes(term, beta, fill = factor(topic))) +
  geom_col(show.legend = FALSE) +
  facet_wrap(~ topic, scales = "free") +
  coord_flip()
```



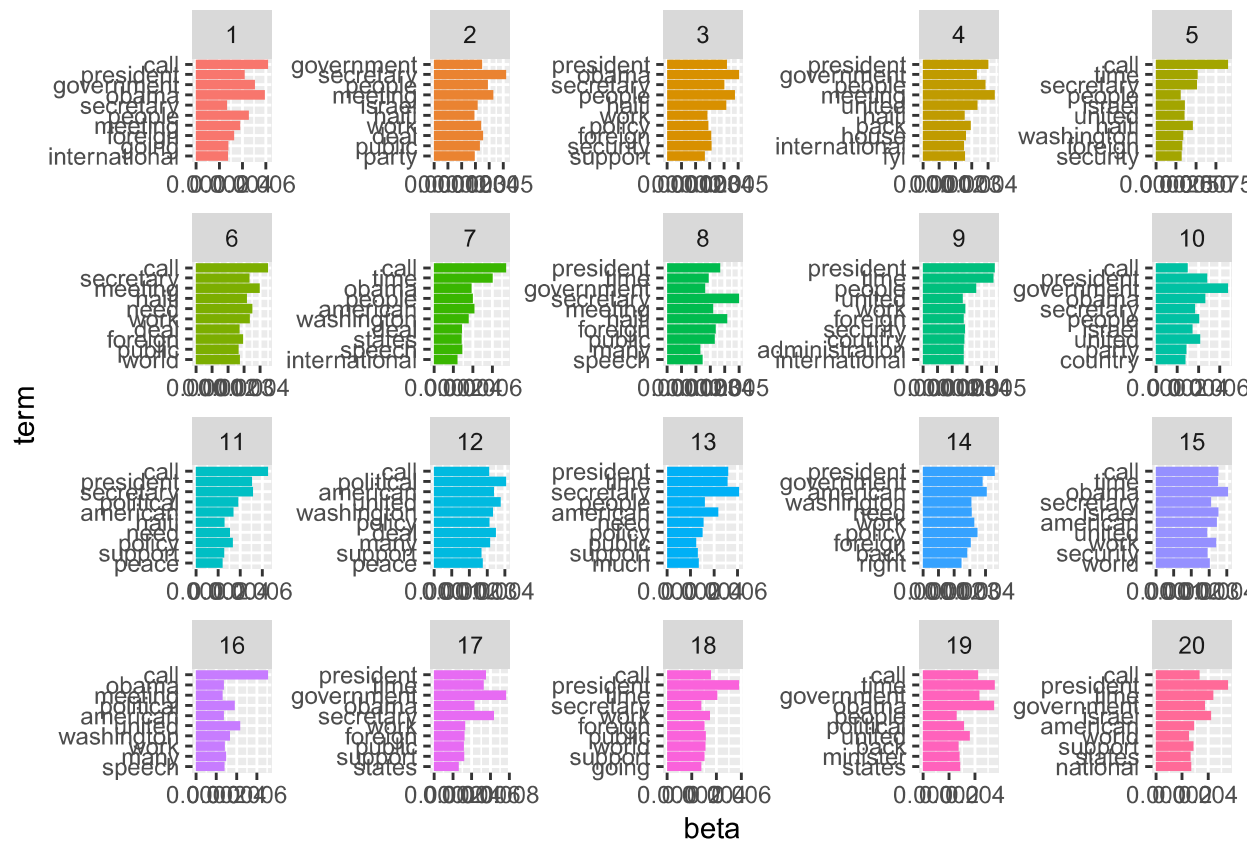
```
# 20

email_nonames_20topic <- LDA(dtm_nonames, k=20)

topics20 <- tidy(email_nonames_20topic, matrix= "beta")

top_terms <- topics20%>%
  group_by(topic)%>%
  top_n(10, beta) %>%
  ungroup()%>%
  arrange(topic, -beta)

top_terms %>%
  mutate(term = reorder(term, beta)) %>%
  ggplot(aes(term, beta, fill = factor(topic))) +
  geom_col(show.legend = FALSE) +
  facet_wrap(~ topic, scales = "free") +
  coord_flip()
```

```
# 30

email_nonames_30topic <- LDA(dtm_nonames, k=30)

topics30 <- tidy(email_nonames_30topic, matrix= "beta")

top_terms <- topics30%>%
  group_by(topic)%>%
  top_n(10, beta) %>%
  ungroup()%>%
  arrange(topic, -beta)

top_terms %>%
  mutate(term = reorder(term, beta)) %>%
  ggplot(aes(term, beta, fill = factor(topic))) +
  geom_col(show.legend = FALSE) +
  facet_wrap(~ topic, scales = "free") +
  coord_flip()
```

